

The RBM mutual-information lower bound is fixed by the marginals, not by a learning principle

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Abstract

A recent review [1] (Sec. 3.2 and Fig. 4 there, after Ref. [34] of the review) characterizes a restricted Boltzmann machine (RBM) learner by two numbers for each visible–hidden pair. The first is the covariance $\eta = \text{Cov}(\sigma_z^v, \sigma_z^h)$, which is read from the imaginary part of an out-of-time-order correlator. The second is the mutual information I . The pair (I, η) is shown to lie between analytic bounds $\text{LB}(\eta)$ and $\text{UB}(\eta)$. Trained RBMs sit on the lower bound for every system size and field ratio of transverse-field Ising drivers, and the review reads this as a learning principle: the network would use the least mutual information compatible with the covariance. We point out that a pair of ± 1 units is fixed by three numbers, the two marginals $m_v = \langle \sigma_z^v \rangle$, $m_h = \langle \sigma_z^h \rangle$ and η . In those variables, $\text{LB}(\eta)$ is exactly the case $m_v = m_h = 0$, and it is the minimum of I over the marginals at fixed η . $\text{UB}(\eta)$ is the perfectly correlated pair with $|m_v| = |m_h| = \sqrt{1 - |\eta|}$. The transverse-field Ising driver is $\mathbb{Z}_2$ -symmetric, so a symmetric RBM has zero marginals and sits on LB by construction. We test the consequence that could have failed. With exact enumeration ($n = 6$ visible spins, $\alpha = 1$), a longitudinal field of 0.1–0.3 moves the same learner off LB by up to 0.17 bits, in step with the measured marginals. In the ordered phase, a symmetry-broken and a symmetric RBM of the same target sit off and on LB respectively. The OTOC reading of η is unaffected; we suggest reporting the marginals alongside (I, η) .

1 The bounds in terms of the marginals

For two ± 1 variables s (visible) and t (hidden) with marginals m_v, m_h and covariance $\eta = \langle st \rangle - m_v m_h$, the joint distribution is

$$p(s, t) = \frac{1}{4} [1 + s m_v + t m_h + st (\eta + m_v m_h)], \quad (1)$$

and the mutual information (in bits) is

$$I(m_v, m_h, \eta) = h\left(\frac{1+m_v}{2}\right) + h\left(\frac{1+m_h}{2}\right) - H(p), \quad (2)$$

with h the binary entropy and $H(p)$ the entropy of Eq. (1). With $\ell(x) = -x \log_2 x$, the bounds of Ref. [1] read $\text{LB}(\eta) = 2 - 2\ell(\frac{1+\eta}{4}) - 2\ell(\frac{1-\eta}{4})$ and $\text{UB}(\eta) = \ell(\frac{1}{2} + \frac{1}{2}\sqrt{1-|\eta|}) + \ell(\frac{1}{2} - \frac{1}{2}\sqrt{1-|\eta|})$. Base-2 logarithms reproduce the plotted endpoints $\text{LB}(0) = 0$ and $\text{LB}(1) = \text{UB}(1) = 1$.

Proposition 1. (i) $\text{LB}(\eta) = I(0, 0, \eta)$. (ii) $\text{UB}(\eta) = I(r, r, \eta)$ with $r = \sqrt{1 - |\eta|}$ (sign of m_h following that of η), which is the locked pair $t = \pm s$. (iii) To leading order in η and the marginals,

$$I - \text{LB}(\eta) \simeq \frac{\eta^2 (m_v^2 + m_h^2)}{2 \ln 2}. \quad (3)$$

Table 1: Trained RBMs (median over 3 seeds, 36 pairs each). Fidelity is with the exact ground state; gaps are in bits.

$g = B/J$	h_z	fidelity	$\langle m_v \rangle$	mean $I - \text{LB}$	max $I - \text{LB}$
1.0	0	1.0000	0.0001	3×10^{-7}	1×10^{-5}
2.0	0	1.0000	0.0000	7×10^{-7}	2×10^{-5}
1.0	0.1	1.0000	0.39	1.6×10^{-2}	0.10
1.0	0.3	1.0000	0.52	1.4×10^{-2}	0.17
2.0	0.1	1.0000	0.07	5.2×10^{-4}	5×10^{-3}
2.0	0.3	1.0000	0.18	2.7×10^{-3}	3×10^{-2}
0.5	0 (random start)	0.50	0.81	1.1×10^{-2}	0.12
0.5	0 ($a = b = 0$ start)	1.0000	0.0002	1.5×10^{-8}	1×10^{-7}

(i) and (ii) follow by substitution into Eq. (1). For (iii), expand I in the correlation coefficient $\rho = \eta / \sqrt{(1 - m_v^2)(1 - m_h^2)}$, using $I \simeq \rho^2 / (2 \ln 2)$. Numerically, on a grid of marginals, $\text{LB}(\eta)$ is the minimum of I at fixed η and $\text{UB}(\eta)$ its supremum. Eq. (3) is 5% low at $\eta = 0.1$ and 12% low at $\eta = 0.3$. The whole (I, η) region is therefore swept by the marginals: at fixed covariance, biased units carry more mutual information because their variances are smaller (Fig. 1, left).

2 Why trained RBMs sit on the lower bound

The learner’s state is the thermal distribution of $H(X) = \sum a_i s_i + \sum b_j t_j + \sum W_{ij} s_i t_j$. The driver $H_{\text{drv}} = -B \sum_i X_i - J \sum_{\langle ij \rangle} Z_i Z_j$ commutes with $\prod_i X_i$, so its ground state has $\langle Z_i \rangle = 0$. An RBM with $a = b = 0$ is invariant under $(s, t) \rightarrow (-s, -t)$, so all its visible and hidden marginals vanish, and by (i) every pair lies exactly on $\text{LB}(\eta)$. Saturation of the lower bound is therefore what $\mathbb{Z}_2$ symmetry implies. It is not evidence that training minimizes mutual information.

This reading makes a prediction that can fail. Break the symmetry, and the points must leave LB by the amount Eq. (3) sets from the measured marginals.

3 Test

We trained RBM wavefunctions $\psi(v) = e^{a \cdot v} \prod_j 2 \cosh(b_j + W_j \cdot v)$ [2] on the driver with an added longitudinal field $-h_z \sum_i Z_i$. The setup: a periodic chain of $n = 6$, $\alpha = 1$, real positive amplitudes (sufficient for this stoquastic driver), exact energy by full enumeration, L-BFGS, and 3 seeds. For every visible–hidden pair of $P(v, h) \propto e^{a \cdot v + b \cdot h + v^\top W h}$ we computed (m_v, m_h, η, I) exactly (Table 1, Fig. 1, right).

- *Symmetric driver.* The review’s observation is reproduced: trained RBMs sit on LB, with gaps below 2×10^{-5} bits, because their marginals vanish.
- *Broken symmetry.* A field of 0.1–0.3 moves the same learner off LB by 5×10^{-4} to 2×10^{-2} bits on average, and up to 0.17 bits for a pair. At $g = 2$, where the marginals are small, Eq. (3) matches the mean gap (5.3×10^{-4} vs 5.2×10^{-4} , and 2.6×10^{-3} vs 2.7×10^{-3}). It underestimates the gap by up to $2\times$ when the marginals are large.
- *Same physics, two positions.* In the ordered phase ($g = 0.5$) a random start breaks the symmetry: the fidelity with the symmetric ground state is 0.50 and the marginals are 0.8, and the points leave LB. A start from $a = b = 0$ keeps the symmetry, finds the symmetric ground state exactly, and the points sit on LB. The target is the same; the position in the (I, η) plane is set by the marginals.

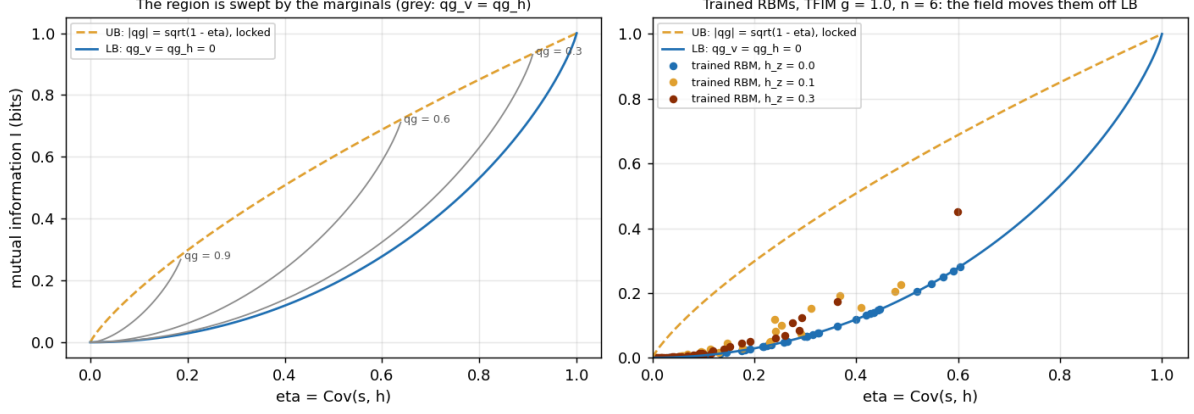


Figure 1: Left: the (I, η) region with the bounds of Ref. [1]; grey curves have fixed marginals $m_v = m_h$. Right: all visible–hidden pairs of trained RBMs at $g = 1$ for three longitudinal fields.

4 Discussion

The OTOC route to η in Ref. [1] and the (I, η) picture remain useful. Our point concerns the interpretation. Lower-bound saturation is the zero-marginal case, and it is fixed by the symmetry of the driver, or broken by the learner or by a field. It does not require a minimum-information principle. A simple way to separate the two readings in larger simulations is to report the marginals $\langle \sigma_z^v \rangle$ and $\langle \sigma_z^h \rangle$ alongside (I, η) , or equivalently the distance to LB predicted by Eq. (3). A learner that minimized mutual information would stay on LB even with $h_z \neq 0$; the RBMs here do not.

Limitations. Our test uses small n with exact enumeration and real RBMs. The stochastic-reconfiguration training with Monte Carlo sampling at larger N used in the review, and its long-range driver, are not reproduced. The claim is limited to the one above.

Code availability. `examples/rbm_mutual_information_qg.py` and its regression tests at <https://github.com/Viny2030/qang> (MIT license), Sec. 47 of `RESEARCH_NOTES.md`. In the notation of that repository, m_v and m_h are the polar biases qq_Z of the two units, and $h((1 + qq_Z)/2)$ is their qg_S [3].

References

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